

FINAL APPROACH

Game Concept Document - Assignment 1

DMET 909 - Video Game Design

Parts 1 and 2: Market Research and Game Concept

Working Genre: Physics-Based Arcade Sports / Precision Sports

Target Platform: PC

Concept: A focused javelin-throw game in which sprint rhythm, release timing, throw angle, stability, and Unity Rigidbody physics combine to determine each throw.

Prepared for academic submission based on the Assignment 1 brief and the supplied DMET 909 Game Concept Document structure.

Document Overview

This document answers Part 1 and Part 2 of Assignment 1. Part 1 establishes the genre, target audience, competitor landscape, and market opportunity. Part 2 translates that research into the game concept and system design for Final Approach.

Part	Coverage
Part 1 - Market Research	Genre selection and motivation; target audience; competitor analysis; market gap.
Part 2 - Game Concept	Working title; concept statement; genre; target audience; USP; player experience/POV; visual/audio style; world fiction; monetization; platform/technology/scope; core loop/mechanics; objectives/progression; systems/interactivity.

Design principle: The project deliberately prioritizes depth in one athletics discipline rather than offering many shallow events. The prototype is intended to be understandable within seconds, but difficult to master consistently.

Part 1 - Market Research and Concept Direction

1. Selecting a Game Genre

Selected genre: Physics-based arcade sports / precision sports, focused on the javelin throw.

Final Approach combines the immediate readability of an arcade sports game with a deeper skill model based on timing and physics. The player builds run-up speed, prepares the throw, selects a launch angle, releases before the foul line, and then observes a physically simulated javelin trajectory. The game remains accessible because the objective is simple and each attempt is short, but the final result depends on several interacting variables rather than one button press.

Track-and-field video games have historically often been packaged as collections of short athletic events. Current and recent competitors continue this structure: Olympic Games Tokyo 2020: The Official Video Game contains more than 15 Olympic events across several platforms [1], Mario & Sonic at the Olympic Games Tokyo 2020 presents a wide collection of accessible Olympic minigames on Nintendo Switch [2], and Summer Games Challenge offers 14 sports disciplines including javelin [5].

This genre was selected because it supports a clear relationship between player input and measurable performance. The concept also evolves an earlier long-jump prototype idea in which player speed affected distance. In Final Approach, the same basic idea is expanded into a richer system: sprint rhythm influences run speed, speed influences stability and launch potential, release timing changes throw quality, launch angle changes trajectory, and the RigidBody simulation produces the final landing distance.

The design goal is therefore not to simulate every detail of professional athletics. It is to create a responsive, understandable sports system in which the player can explain why a throw succeeded or failed and can intentionally improve on the next attempt.

2. Target Audience

The primary target audience is players who enjoy short-session sports games, score chasing, timing mechanics, and physics-based mastery. The intended core audience is approximately 13-35 years old, including teenagers, university-age players, and adults who enjoy accessible competition without requiring the complexity of a full sports simulator.

The audience is expected to include players interested in athletics and Olympic sports, casual sports games, timing/rhythm challenges, physics-based games, personal-best systems, and repeated skill improvement. The game is designed primarily for PC and for short sessions in which several attempts can be completed quickly.

Player motivation is centered on mastery. Instead of relying on character-stat grinding, the player improves by learning the timing and interaction of the game systems. This direction is supported by player feedback around Athletics Mania: Track & Field, where one highly detailed App Store review praised the need to maintain a good running tempo rather than simply spam the run input [3]. The same source also documents user complaints about bugs and friction, showing the importance of reliable feedback and clean execution.

The intended emotional pattern is: preparation -> rhythm -> tension near the line -> release -> anticipation during flight -> measurable result -> immediate desire to retry. The game therefore targets players who enjoy improving a personal score through practice rather than players looking primarily for a narrative campaign.

3. Competitor Analysis

The competitor analysis focuses on games that are relevant because they contain track-and-field events, javelin, or short arcade sports challenges. The objective is not to claim that the market has no javelin games; instead, it is to identify how existing games structure their athletics experiences and where Final Approach can differentiate.

Competitor	Platforms / Presence	Strengths	Relevant Limitations / Opportunity
Olympic Games Tokyo 2020: The Official Video Game	PS4, Switch, PC, Xbox One, Xbox Series X [1]	More than 15 events; accessible; online multiplayer; recognizable Olympic format.	PC critics described mechanics as basic and long-term solo replay value as limited [6]. Opportunity: greater depth in one event.
Mario & Sonic at the Olympic Games Tokyo 2020	Nintendo Switch [2]	Strong presentation; broad event variety; family/party accessibility.	Reviews emphasize casual, bite-sized play rather than deep simulation [2]. Opportunity: preserve accessibility while increasing mastery.
Athletics Mania: Track & Field	Mobile / App Store; includes javelin [3][4]	Large athletics selection; career/RPG systems; rhythm-sensitive running.	User reviews mention bugs and friction; timing clarity is important. Opportunity: transparent technique feedback and cleaner single-event focus.
Summer Games Challenge	PC / Steam [5]	14 disciplines including javelin; single-player and local multiplayer.	Broad multi-event focus; very small Steam review sample. Opportunity: specialization and deeper javelin mechanics.
Summer Athletics: The Ultimate Challenge	PC, PS2, Wii, Xbox 360 [7]	Large event collection; straightforward short-event structure.	GameSpot criticized repetitive, shallow events and unclear system feedback [7]. Opportunity: replace button-mashing with interdependent mechanics.

3.1 Key Competitor Findings

Olympic Games Tokyo 2020. Metacritic lists the title on PlayStation 4, Nintendo Switch, PC, Xbox One, and Xbox Series X, with more than 15 Olympic events and online multiplayer [1]. On PC, the critic aggregate is 56, and the displayed review excerpts repeatedly question long-term replay value and mechanical depth [6]. This makes it an important example of the trade-off between event breadth and event depth.

Mario & Sonic at the Olympic Games Tokyo 2020. The title has a 69 Metascore on Nintendo Switch. Critic summaries praise the variety and approachable party-game structure, while also characterizing the experience as casual and bite-sized rather than deeply simulated [2]. The useful lesson is that accessibility and presentation are valuable, but Final Approach needs a stronger mastery loop.

Athletics Mania: Track & Field. The App Store description includes javelin alongside sprints, jumps, and other throwing events [4]. Its U.S. review page currently shows a 4.5/5 aggregate from 1.8K ratings and includes a detailed review praising tempo-based running while also noting major bugs [3]. This competitor supports the idea that players can appreciate timing and rhythm systems when they are readable and responsive.

Summer Games Challenge. Steam lists 14 disciplines, including javelin, and supports single-player and local multiplayer for up to four players [5]. The Steam page currently exposes only three user reviews, so it is not appropriate to draw strong conclusions about broad player reception from that sample. It remains useful as evidence of a recent PC athletics product that still uses the multi-event package structure.

Summer Athletics: The Ultimate Challenge. GameSpot scored the title 3/10 and criticized the majority of events as dull, mechanically similar, and reliant on old button-mashing conventions.

The review also highlights unclear feedback in some systems [7]. For Final Approach, this reinforces the need for meaningful variables and understandable performance feedback.

3.2 Market Gap and Opportunity

The market research suggests an opportunity for a focused PC athletics game that treats one throwing discipline as a deeper skill-based physics experience rather than one short event inside a large Olympic collection. This is a positioning opportunity, not a claim that no other javelin games exist.

- Single-event depth: concentrate development effort on one event so sprint rhythm, release timing, angle, stability, foul rules, and projectile physics can meaningfully interact.
- Physics-driven outcomes: derive launch conditions from player performance and allow Unity physics to produce the resulting trajectory and landing point.
- Transparent feedback: show speed, angle, release quality, stability, and result information so weak performance feels explainable rather than arbitrary.
- Low barrier, high mastery: keep the control set compact while making consistent high-performance throws require practice.
- Short-session replayability: use three-attempt rounds, personal bests, and immediate reset to support repeated improvement without grind.

The resulting positioning statement is:

Final Approach is a physics-based arcade javelin game that replaces the shallow one-event minigame structure common in multi-sport athletics titles with a focused mastery loop built around sprint rhythm, release timing, launch angle, stability, and physically simulated projectile motion.

Part 2 - Game Concept Document

1. Introduction

Working title: *Final Approach*. A single-player javelin game built around the final approach to the foul line, when an athlete must convert running speed, stability and precise release timing into a legal throw.

2. Game Description

Concept statement. *Build sprint rhythm, select a throw angle and release the javelin as close to the foul line as possible without crossing it. Unity physics determines its flight and landing distance.*

Each short attempt ends with a measured result and feedback that helps the player adjust the next throw. A round contains three attempts; the best legal distance is the round score.

3. Key Features

- **Focused javelin event:** deeper skill interactions instead of a collection of unrelated athletics minigames.
- **Rhythm-based run-up:** timed Space presses build speed; excessive tapping is less effective.
- **Risk-versus-control release:** stronger timing near the foul line, with a foul if released beyond it.
- **Physics-based flight:** angle, launch speed and stability shape the Rigidbody trajectory.
- **Readable improvement:** speed, angle, release quality, distance and personal best support replayability.

4. Genre and Target Audience

Primary genre: arcade sports. **Secondary genre:** physics-based precision/skill. Intended for PC players approximately aged 13-35 who enjoy athletics, timing mechanics, short score-chasing sessions and improving through practice. The accessible control set does not require prior javelin knowledge.

5. Platforms, Technology and Scope

Platform and input	PC desktop; keyboard; single-player.
Engine and camera	Unity 6 Universal Render Pipeline (URP); static/manually placed third-person side/elevated camera; no Cinemachine.
Prototype scope	One playable javelin event in an athletics stadium; three-attempt rounds; reusable Prefabs. Assignment minimums: 20 different GameObject models/placeholders, 20 distinct materials/textures and a floor material.

6. Market Analysis (Summary of Part 1)

Part 1 compares multi-event athletics titles, including *Olympic Games Tokyo 2020*, *Mario & Sonic at the Olympic Games Tokyo 2020*, *Athletics Mania* and *Summer Games Challenge* [1]-[7]. The documented opportunity is to emphasize **single-event mastery, visible technique feedback and physics-driven results** while retaining accessible, short-form play. See Part 1 for competitor strengths, limitations, reviews and source details.

Part 2 - Game Concept Document (continued)

7. Core Gameplay Loop and Controls

Start attempt → build speed → approach line → prepare and aim → release → observe flight → record distance → review and retry.

Space	Tap rhythmically to accelerate from a base of 3 m/s toward a 10 m/s cap.
Left Shift	Hold to prepare; release to launch before the foul line.
W / S	Adjust throw angle (15-60 degrees; default 35 degrees).
Enter / H	Continue to the next attempt / open or close How to Play.

8. Mechanics, Rules and Scoring

Release quality improves closer to the line: **Early 0.70** (>3 m), **Safe 0.85** (3-1.5 m), **Good 0.95** (1.5-0.5 m), **Perfect 1.00** (0.5-0 m); crossing before release is a foul. Stability decreases slightly at high speed (1.00 up to 8 m/s; 0.97 at 8-9; 0.93 at 9-10).

Prototype launch-speed tuning: **(18 + run speed x 0.9) x release quality x stability**. A Rigidbody handles gravity and collision. The first valid landing measures distance in metres from the line. Fouls receive no distance; the best legal result across three attempts wins the round.

9. Player Experience and Point of View

The player is a competitive javelin athlete viewed by a readable third-person side/elevated camera. The desired arc is **preparation → rhythm → tension → release → anticipation → result → retry**. Short attempts and transparent feedback reinforce skill-based learning.

10. Game World, Visual and Audio Style

A fictional modern athletics circuit provides a light competition context, with one stadium for the prototype. The stylized-realistic runway, foul line, grass landing field, distance markers, stadium seating and equipment make the play space legible. A sports-broadcast-inspired HUD prioritizes speed, angle, attempts and result. Crowd ambience and distinct preparation, release, landing and foul cues support the actions.

11. Objectives, Progression and Systems

The immediate objective is the longest valid throw and a new personal best. Progression comes from improved sprint rhythm, timing, angle choice and consistency, not purchased stats. The prototype connects run-up, angle, release, stability, projectile physics, landing measurement, attempt tracking and UI feedback in one repeatable state flow.

12. Monetization and Future Scope

The academic prototype has **no monetization**. A possible full release would use a single premium purchase without pay-to-win. Optional later additions include new venues, tougher timing windows, light wind, cosmetic personalization and competitions; online multiplayer, career systems and complex weather remain outside the MVP.

References

- [1] Metacritic. Olympic Games Tokyo 2020: The Official Video Game - reviews/details. Platforms, release information, event breadth, aggregate review data.
<https://www.metacritic.com/game/olympic-games-tokyo-2020-the-official-video-game/>
- [2] Metacritic. Mario & Sonic at the Olympic Games Tokyo 2020 - critic reviews. Review aggregate and critic commentary on variety, accessibility, and casual/bite-sized play.
<https://www.metacritic.com/game/mario-and-sonic-at-the-olympic-games-tokyo-2020-2019/critic-reviews/>
- [3] Apple App Store (US). Athletics Mania: Track & Field - Ratings & Reviews. Current rating page and player comments including tempo-sensitive running and reported bugs.
<https://apps.apple.com/us/app/athletics-mania-track-field/id1464651738?platform=iphone&see-all;=reviews>
- [4] Apple App Store. Athletics Mania: Track & Field listing. Discipline list includes javelin, jumps, running events, and other throws; multiplayer/RPG/career features.
<https://apps.apple.com/eg/app/athletics-mania-track-field/id1464651738>
- [5] Steam. Summer Games Challenge. Product page listing 14 disciplines including javelin, single-player/local multiplayer, and current review count. <https://store.steampowered.com/app/3024050/>
- [6] Metacritic. Olympic Games Tokyo 2020: The Official Video Game - PC Critic Reviews. PC aggregate and review excerpts discussing basic mechanics and replay value. <https://www.metacritic.com/game/olympic-games-tokyo-2020-the-official-video-game/critic-reviews/?platform=pc>
- [7] GameSpot. Carolyn Petit, Summer Athletics: The Ultimate Challenge Review, 10 Sep. 2008. Critique of repetitive minigames, rapid-input mechanics, unclear feedback, and presentation.
<https://www.gamespot.com/reviews/summer-athletics-the-ultimate-challenge-review/1900-6197408/>
- [8] DMET 909. Assignment 1: Video Game Concept Document and Prototype Development. Supplied course assignment brief. Used for required sections, prototype constraints, and asset requirements.
- [9] DMET 909. Detailed Game Design Documentation Template. Supplied course template. Used to structure concept, target audience, player experience, systems, world, technology, and progression sections.

Research note: Online review aggregates and storefront review counts can change over time. They are used here as contextual evidence for competitor analysis, not as guarantees of future market performance.